Airframe Technology Master

Fatigue & DT
Lecture 17
Application. Exercises.

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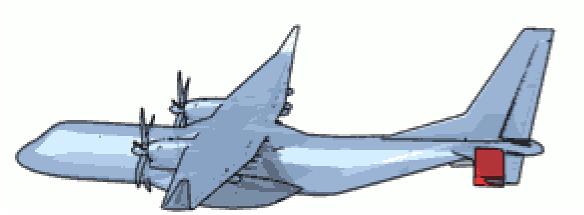
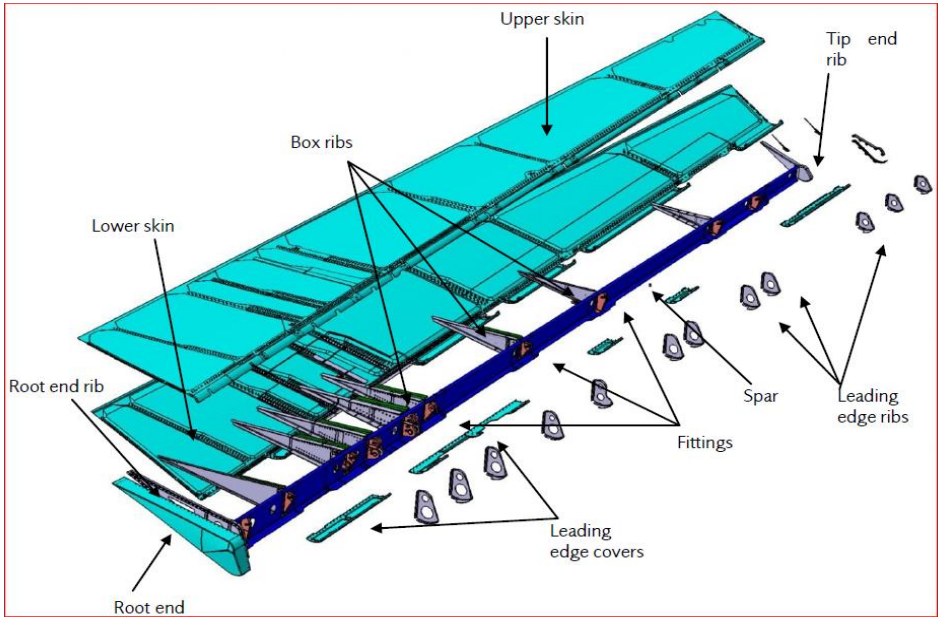
23th October 2025

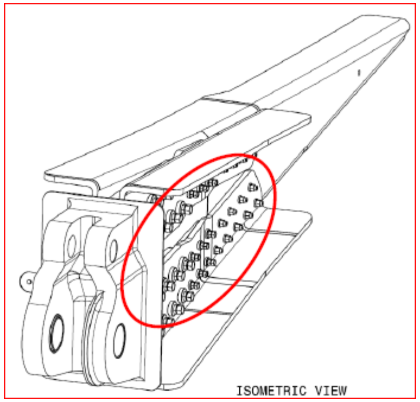
ANALYSIS PROCESS OVERVIEW

How to perform a full fatigue analysis?

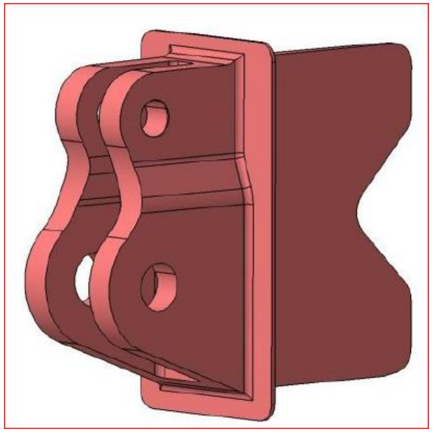


Elevator overview





Elevator fitting assembly



Elevator fitting

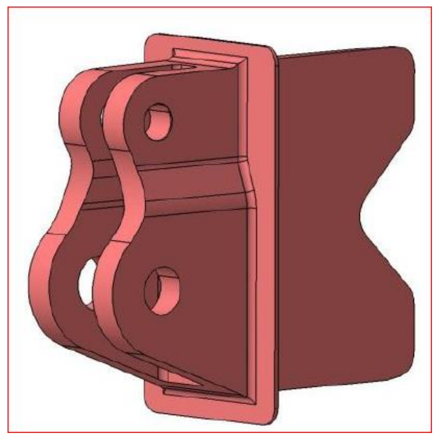
ANALYSIS PROCESS OVERVIEW

Gathering information for the analysis

Which is the main load?
(loading)

How is it?
(geometry)

How does the load vary in time?
(cycles)



What is it made of?
(material)

How does the load distribute within the fitting?
(stress)

Which are the critical areas from fatigue&DT point of view?
(locations)

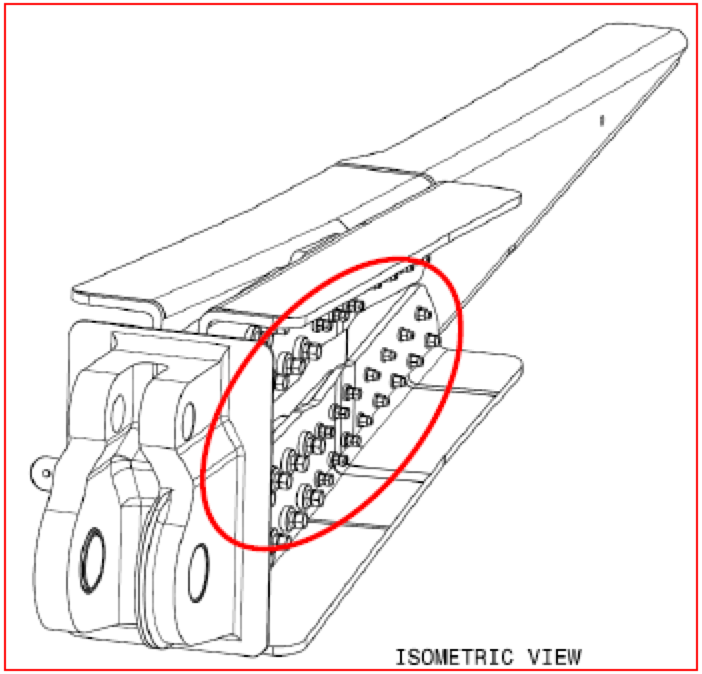
ANALYSIS PROCESS OVERVIEW

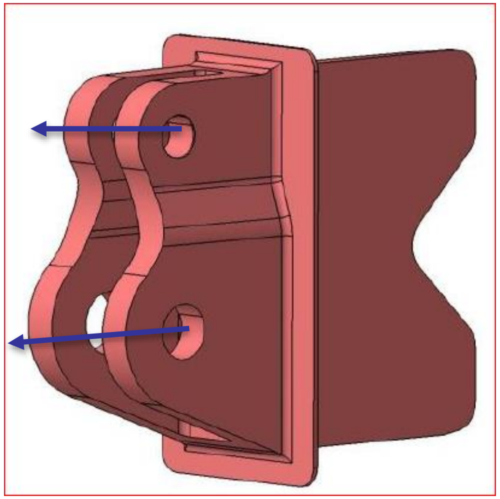
Steps

1. Determine the areas prone to develop fatigue: Locations
2. Gather all the information needed for the analysis: Geometry, loads, material, etc...
3. Think about the best way to make the justification of each location: Can any of the locations be covered by another location? This would save analysis time!
4. Perform fatigue analysis.
5. Perform DT analysis.
6. Prepare the report containing the fatigue and DT analyses.
7. Send the certification document to the authorities, if needed.

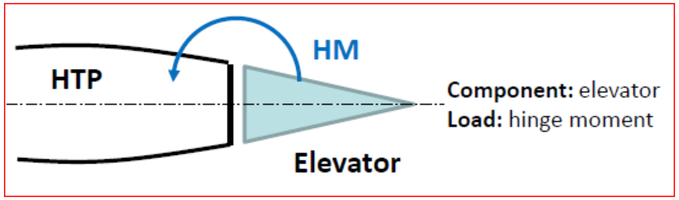
LOADING

Which loads apply to the part to be analysed?



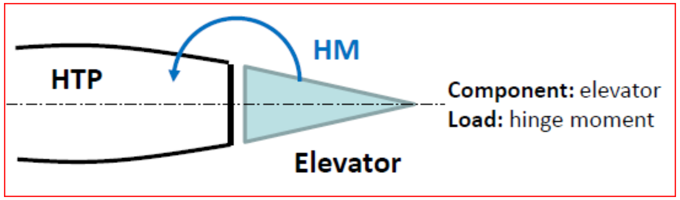


The part to be analysed is the actuator fitting. The loads in the fitting are applied in the lugs: actuator lug and hinge lug (reaction bar).



The main load in the elevator comes from the aerodynamic load which is compensated with the hinge moment at the hinge line.

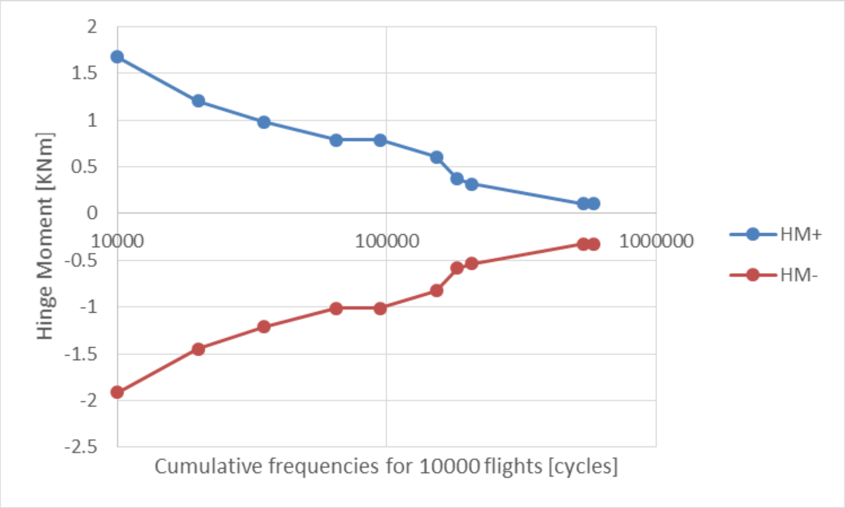
LOAD SPECTRUM



Elevator Angle (+/-) in degrees	Hinge Moment max. in kNm	Hinge Moment min. in kNm	Aerodynamic load max. in kN	Aerodynamic load min. in kN	Occurrences per 10000 flights
5	1,673	-1,911	9,558	-10,920	10000
12	1,207	-1,445	6,897	-8,255	10000
5	0,980	-1,209	5,601	-6,907	15000
3	0,787	-1,010	4,496	-5,771	30000
4	0,785	-1,010	4,488	-5,771	29250
2	0,605	-0,824	3,459	-4,707	58500
1	0,369	-0,585	2,110	-3,345	30000
3	0,315	-0,533	1,798	-3,046	25000
1	0,105	-0,321	0,603	-1,833	331030
1	0,105	-0,321	0,603	-1,833	50000

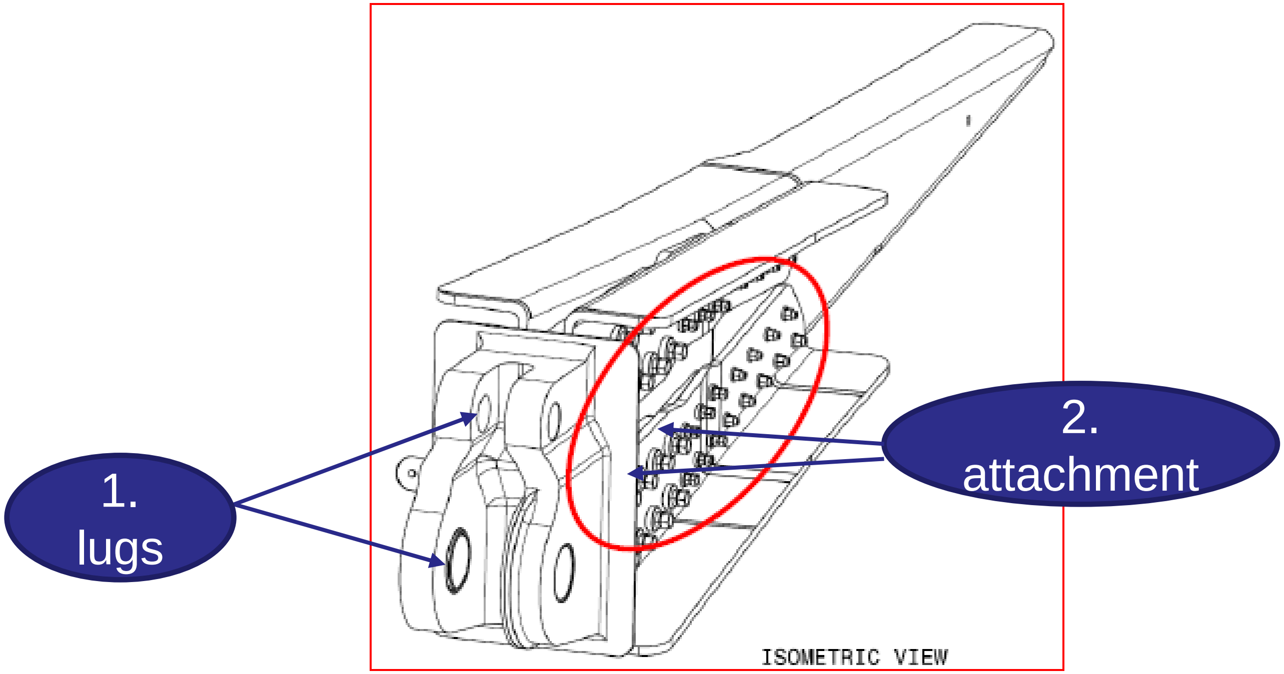
DSG=10000Flights

The spectrum is given in hinge moment and aerodynamic load.
This spectrum can be converted to a load spectrum on the lugs. The transfer factor can be obtained with a FEM or by hand (with simplifications).



LOCATIONS TO BE ANALYSED

Which locations would you select for your analysis?

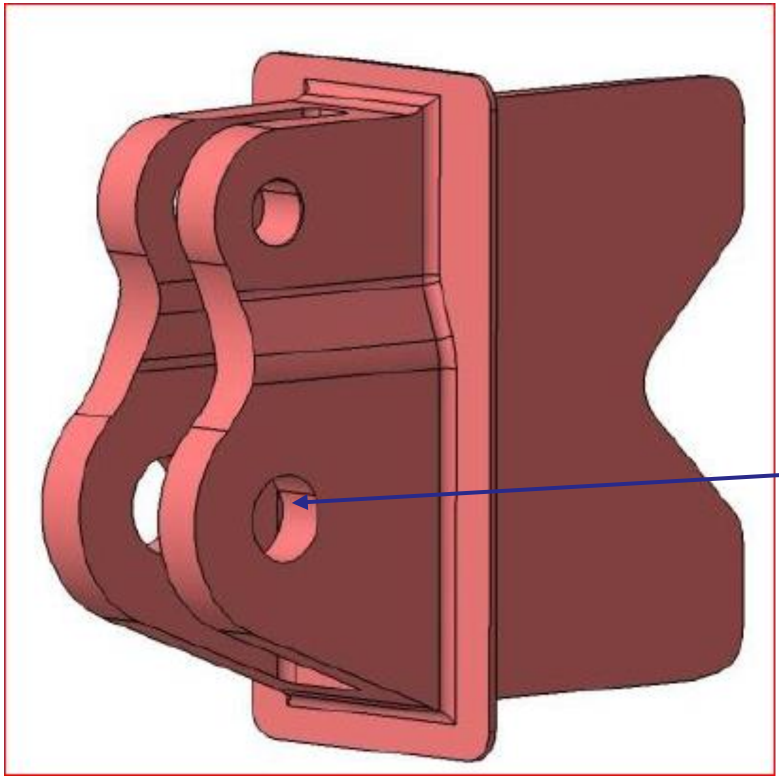


1. Lugs: Load enters in the fitting through the lug
2. Attachment: The load is tranferred from the fitting to the rib

FATIGUE ANALYSIS PROCESS

1. Get the material and geometry for the location
2. Get the Kt and take note of the reference stress
3. Convert the spectrum to the reference stress
4. Enter in the SN curve to get the fatigue life for the location

LUG FATIGUE ANALYSIS



Let's start the fatigue analysis!

Actuator Lug

LUG FATIGUE ANALYSIS

1. Get the material and geometry for the location

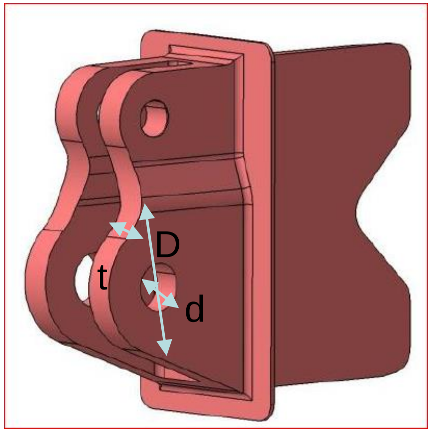
For simplification, the lug will be considered straight lug instead of angled lug.

2. Get the Kt and take note of the reference stress

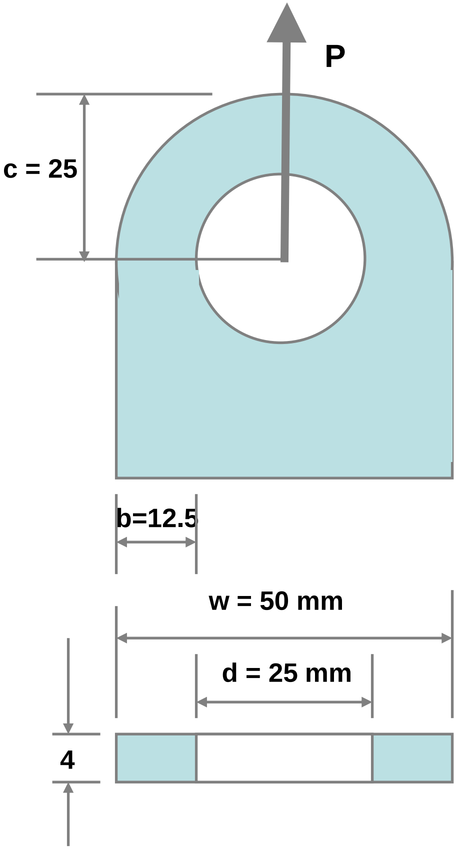
3. Convert the spectrum to the reference stress

4. Enter in the SN curve to get the fatigue life for the location

LUG FATIGUE ANALYSIS



D= 50 mm
d= 25 mm
c= 25 mm
t= 4 mm



Pin material: Steel
Fitting material: Al 7475-T7351

LUG FATIGUE ANALYSIS

1 Get the material and geometry for the location ✓

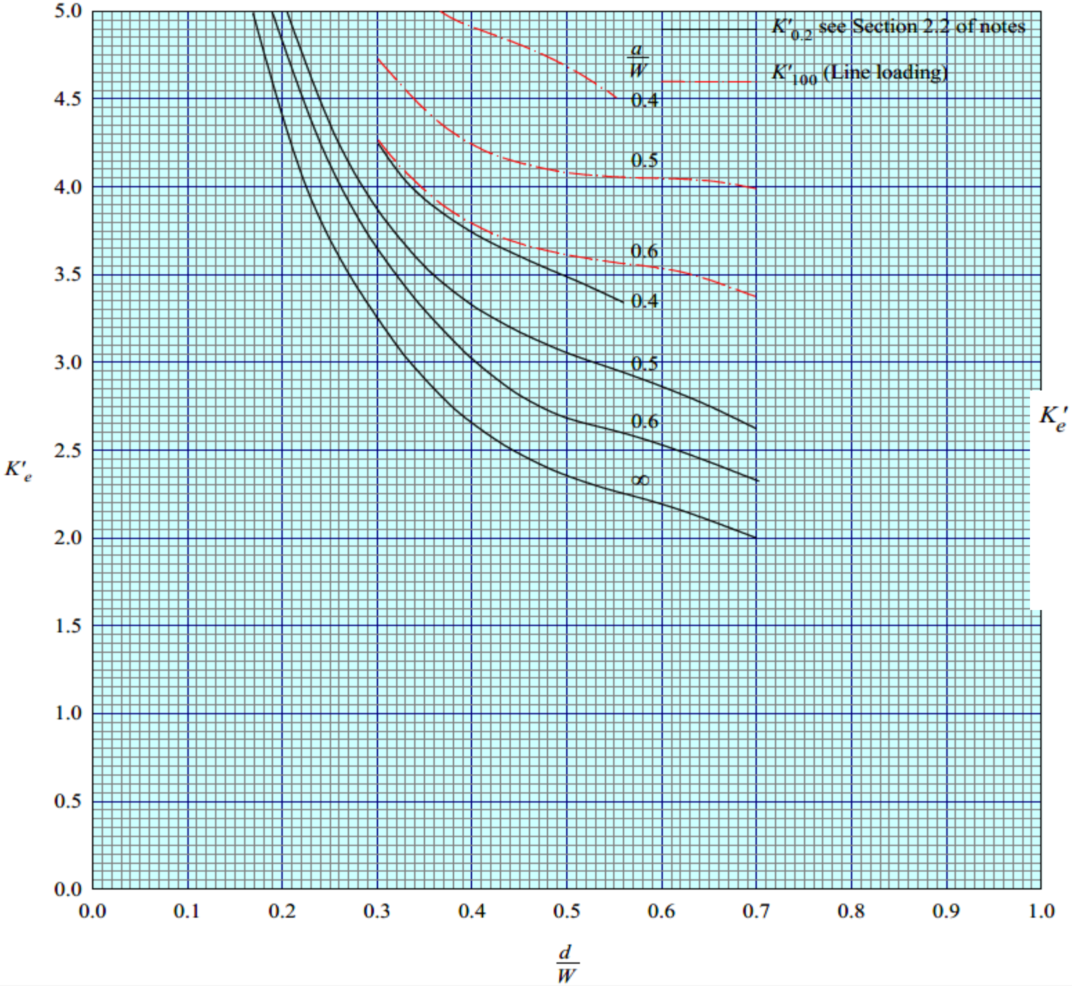
2 Get the Kt and take note of the reference stress

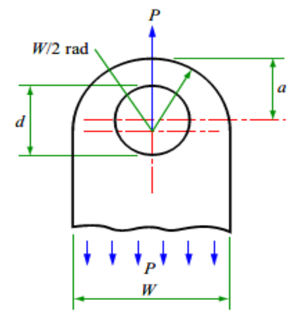
- *Kt lug and Kt pin bending from ESDU 81006*

3 Convert the spectrum to the reference stress

4 Enter in the SN curve to get the fatigue life for the location

LUG FATIGUE ANALYSIS





K'_e stress concentration factor at e per cent clearance for lugs with t/d less than 0.5, defined by

$$f_{max} = K'_e f_{ref}$$

where $$f_{ref} = \frac{P}{(W-d)t}$$

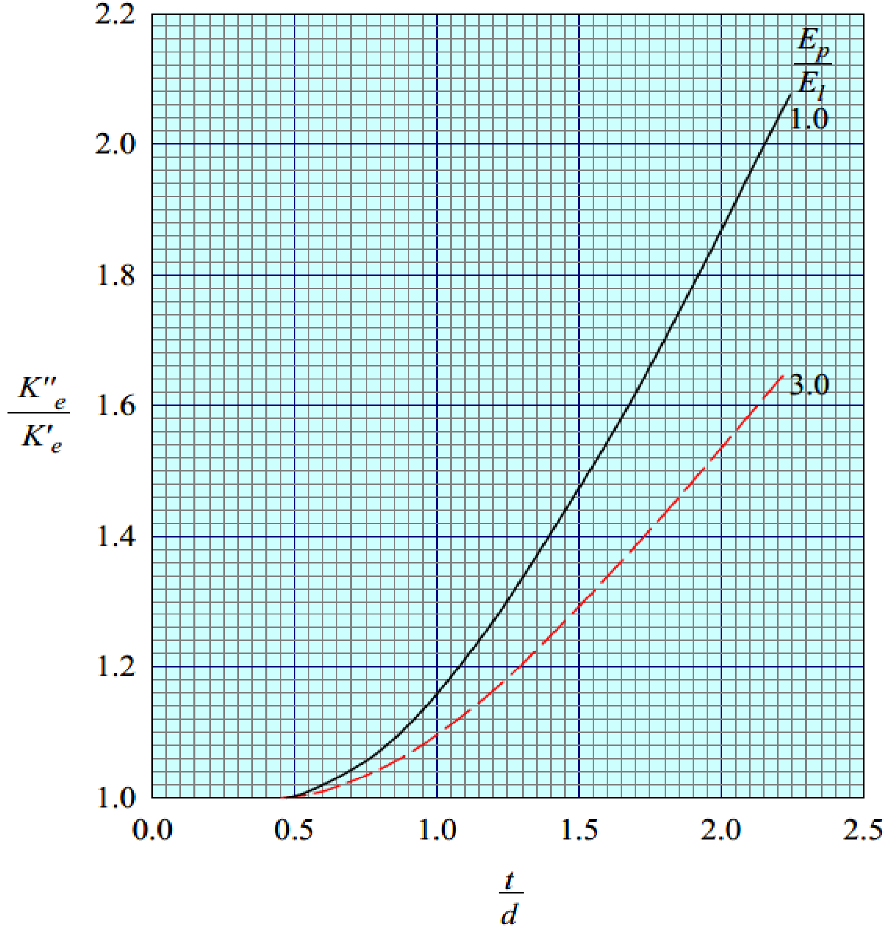


FIGURE 4 EFFECT OF LUG THICKNESS

K''_e value of K'_e when t/d is greater than 0.5 (see Section 2.5)

LUG FATIGUE ANALYSIS

1. Get the material and geometry for the location ✓
2. Get the Kt and take note of the reference stress ✓
3. Convert the spectrum to the reference stress
   - ❑ *Get the transfer factor from hinge moment to load in the lug*
   - ❑ *Get the transfer factor from load in the lug to the reference stress for the Kt*
4. Enter in the SN curve to get the fatigue life for the location

LUG FATIGUE ANALYSIS

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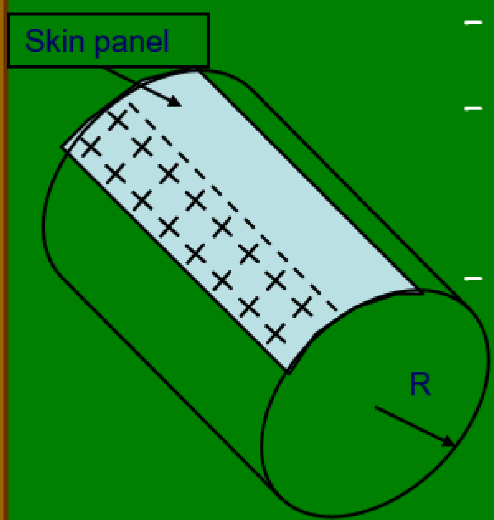
Exercise

REMINDER

Exercise: Determine the fatigue allowable stress for a pressurized fuselage:



- 60,000 FC design service goal,
- Inspection threshold goal= 30,000 FC
- Skin material: Al 2024-T3 clad sheet, thickness=t=2mm, surface roughness= 3.2 micrometers,
- Rivets: aluminium solid rivets installed with clearance fit, Diameter=D=4mm, pitch=p= 5D, rivet edge distance=e=2D, countersunk depth=c= 0.96mm
- Fuselage diameter: R= 2900 mm
- ΔP= 6.0 psi

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Exercise

REMINDER

3. Obtain the relevant loads and stresses:

- Operating load: differential pressure: $\Delta P = 6.0$ psi
- Fatigue limit load*: $1.15\ \Delta P = 6.9$ psi
- Static limit load: $1.33\ \Delta P = 7.98$ psi
- Hoop stress: (far field stress)

$$\sigma = \Delta P R / t$$

$$\sigma_{op} = 6.0\ E\text{-}3 \times 6.895\ MPa \times 2900\ mm / 2mm = 60.0\ MPa$$

$$\sigma_{fat} = 6.9\ E\text{-}3 \times 6.895\ MPa \times 2900\ mm / 2mm = 69.0\ MPa$$

$$\sigma_{sta} = 7.98\ E\text{-}3 \times 6.895\ MPa \times 2900\ mm / 2mm = 79.8\ MPa$$

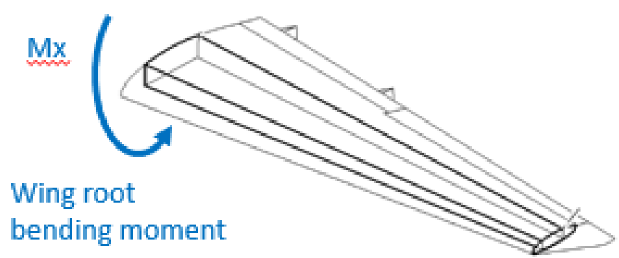
* from CS 25.571 or FAR 25.571

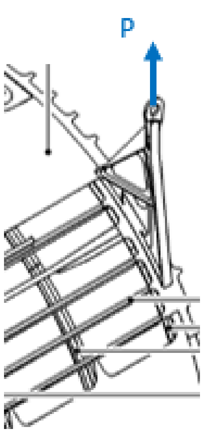
LUG FATIGUE ANALYSIS

Exercise

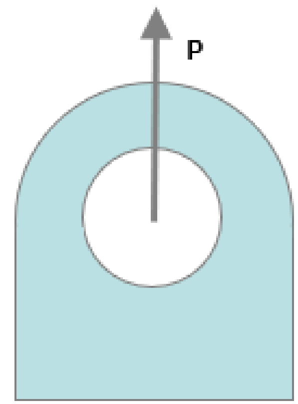
REMINDER

Transfer Function

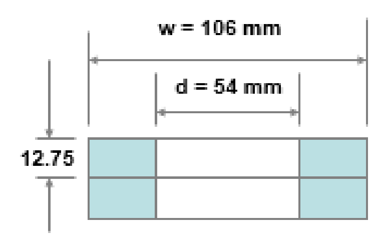




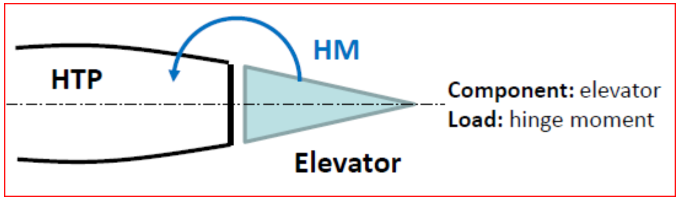
$$Ftr_1 = \frac{P}{Mx} = 0.125$$



$$Ftr_2 = \frac{\sigma_{gross}}{P} = \frac{1}{w \cdot t} = \frac{1}{106 \cdot 25.5}$$



For the analysis we need the load in the actuator lug. This load can be obtained directly from a FEM (as load in CELAS, por example), or by hand.

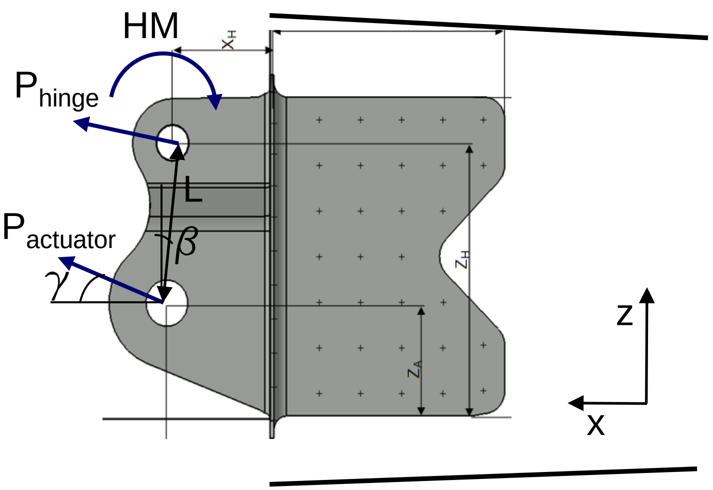


The actuator load is in the bar direction and since there is a bearing, there is no moment in the lug.

Simplification for the exercise:

- elevator deflection δ is small. It is neglected for the load distribution.

L=112 mm, β= 10 deg, γ= 5 deg



$$HM=P_{act\_x}\, L_z + P_{act\_z}\, L_x$$

$$P_{act\_x}=P_{act}\cos\gamma\,; \quad P_{act\_z}=P_{act}\,\mathrm{sen}\gamma$$

$$L_z=L\cos\beta\,; \quad L_x=L\,\mathrm{sen}\beta$$

$$HM=P_{act}\, L\cos\gamma\cos\beta + P_{act}\, L\,\mathrm{sen}\,\gamma\,\mathrm{sen}\beta$$

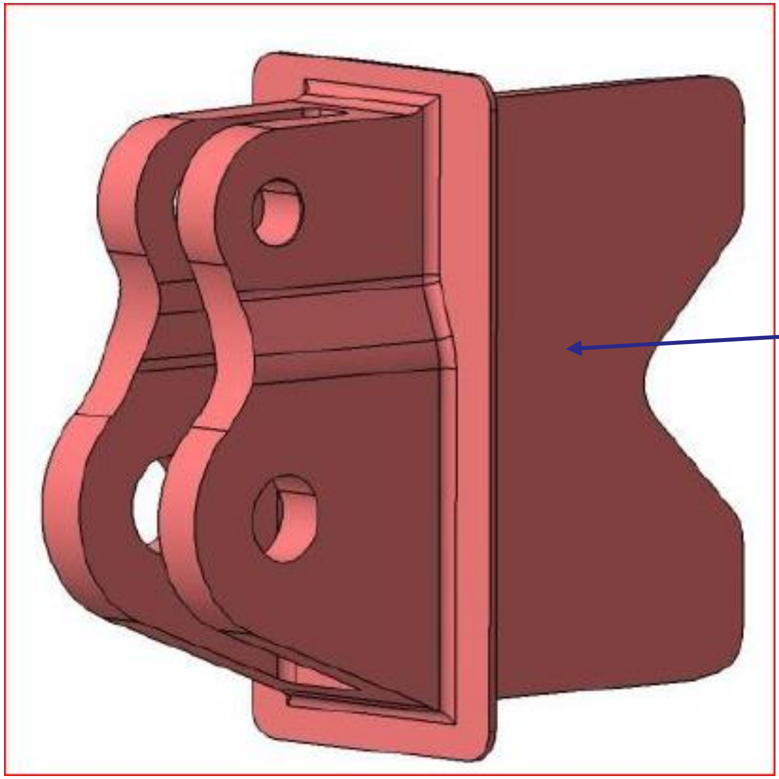
$$P_{act}=HM/[L(\cos\gamma\cos\beta + \mathrm{sen}\,\gamma\,\mathrm{sen}\beta\ )]$$

LUG FATIGUE ANALYSIS

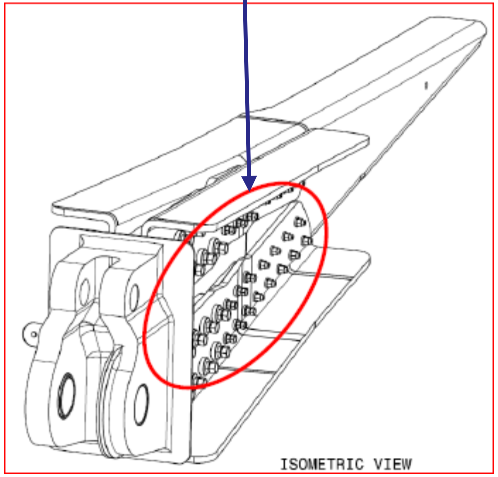
1. Get the material and geometry for the location ✓
2. Get the Kt and take note of the reference stress ✓
3. Convert the spectrum to the reference stress ✓
4. Enter in the SN curve to get the fatigue life for the location

   - *Interpolate if needed*

ATTACHMENT FATIGUE ANALYSIS



Attachment



Let's start the fatigue analysis!

ATTACHMENT FATIGUE ANALYSIS

1. Get the material and geometry for the location

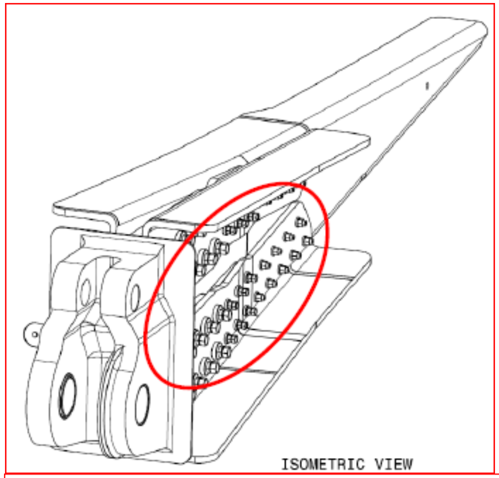
For simplification, single shear joint is considered for the exercise

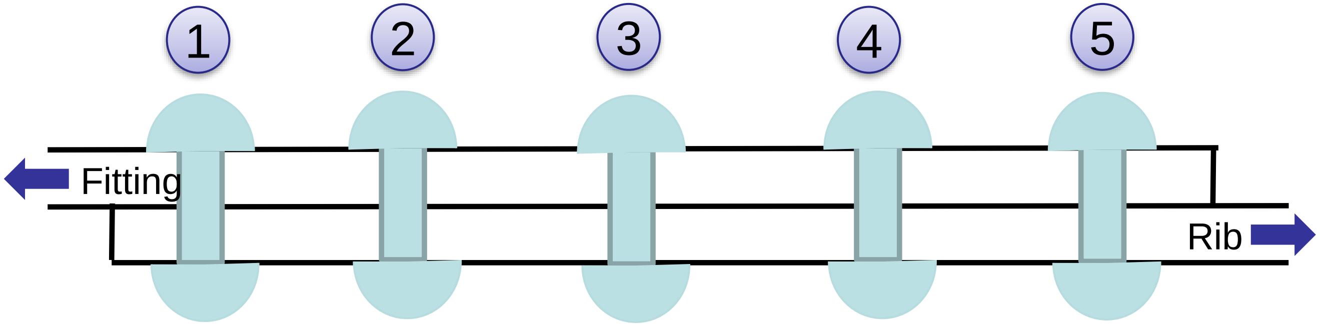
2. Get the fastener loads and stresses at the joint. Obtain the Kt taking a reference stress.

3. Convert the spectrum to the reference stress

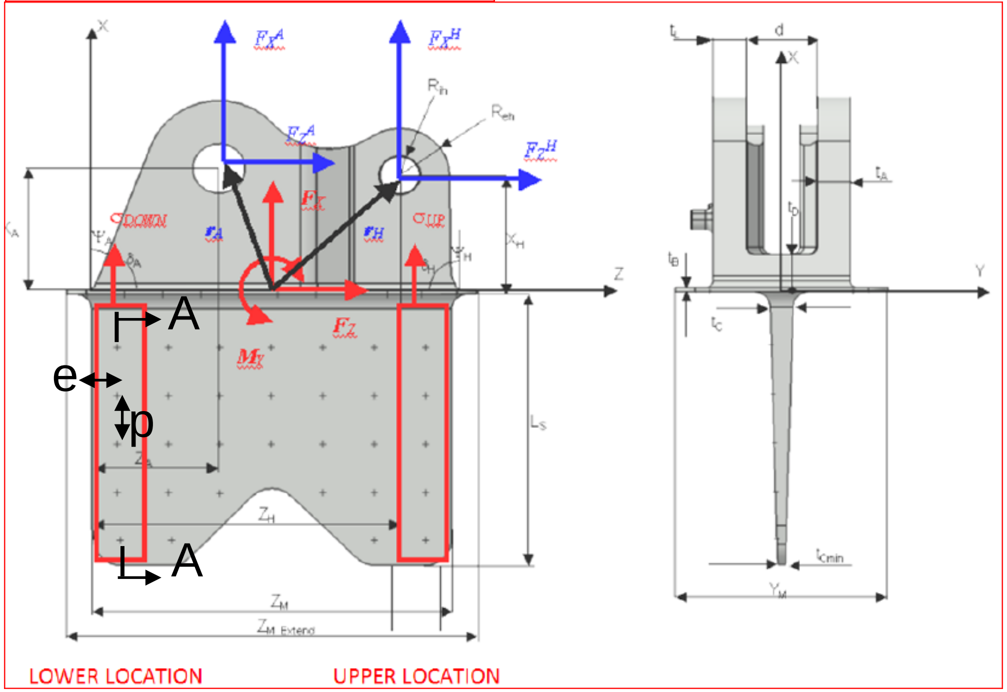
4. Enter in the SN curve to get the fatigue life for the location

ATTACHMENT FATIGUE ANALYSIS





Section A-A



D=4.8 mm
protruding aluminum fasteners
plate and rib: Al 7475-T7351
pitch=p = 4D
edge distance=e= 2D
thickness rib and fitting=t= 1.4 mm

ATTACHMENT FATIGUE ANALYSIS

1. Get the material and geometry for the location ✓

2. Get the fastener loads and stresses at the joint. Obtain the Kt taking a reference stress.
   - *Ktgross, Ktbrg, Ktpb, Ktt*
   - *Sref (typically at the beggining of the load transfer)*

3. Convert the spectrum to the reference stress

4. Enter in the SN curve to get the fatigue life for the location

ATTACHMENT FATIGUE ANALYSIS

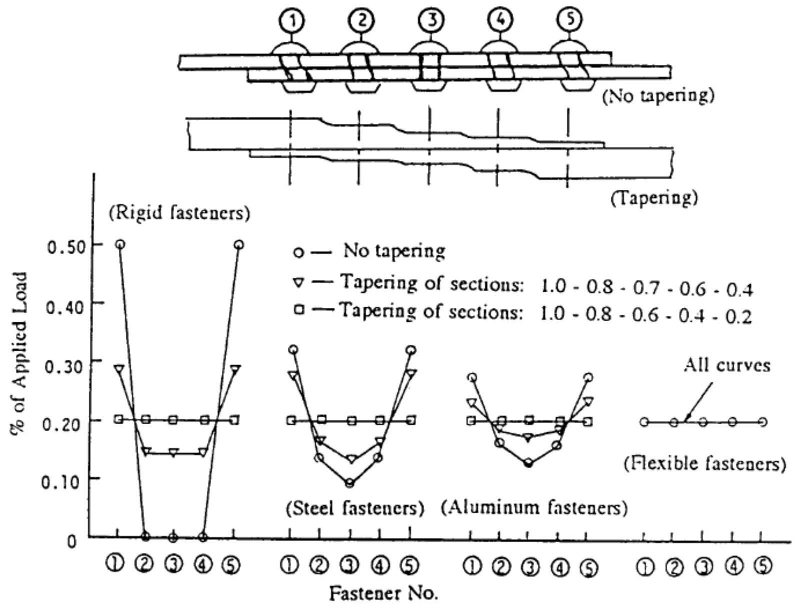


Fig. 9.12.5 Variation of Fastener Load Distribution due to Fastener and Plate Stiffne

The loads in the fasteners are obtained considering the flexibility of the joint, which depends on the plate thickness and material and on the fastener diameter and material.

The more flexible the joint is, the more even the load distribution is.

The load distribution is usually obtained from a FEM or with an appropiate software.

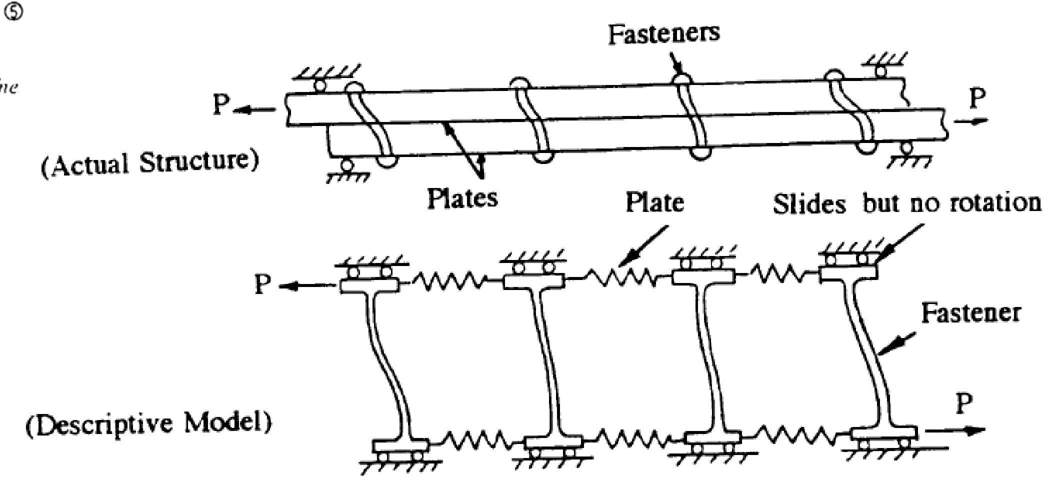
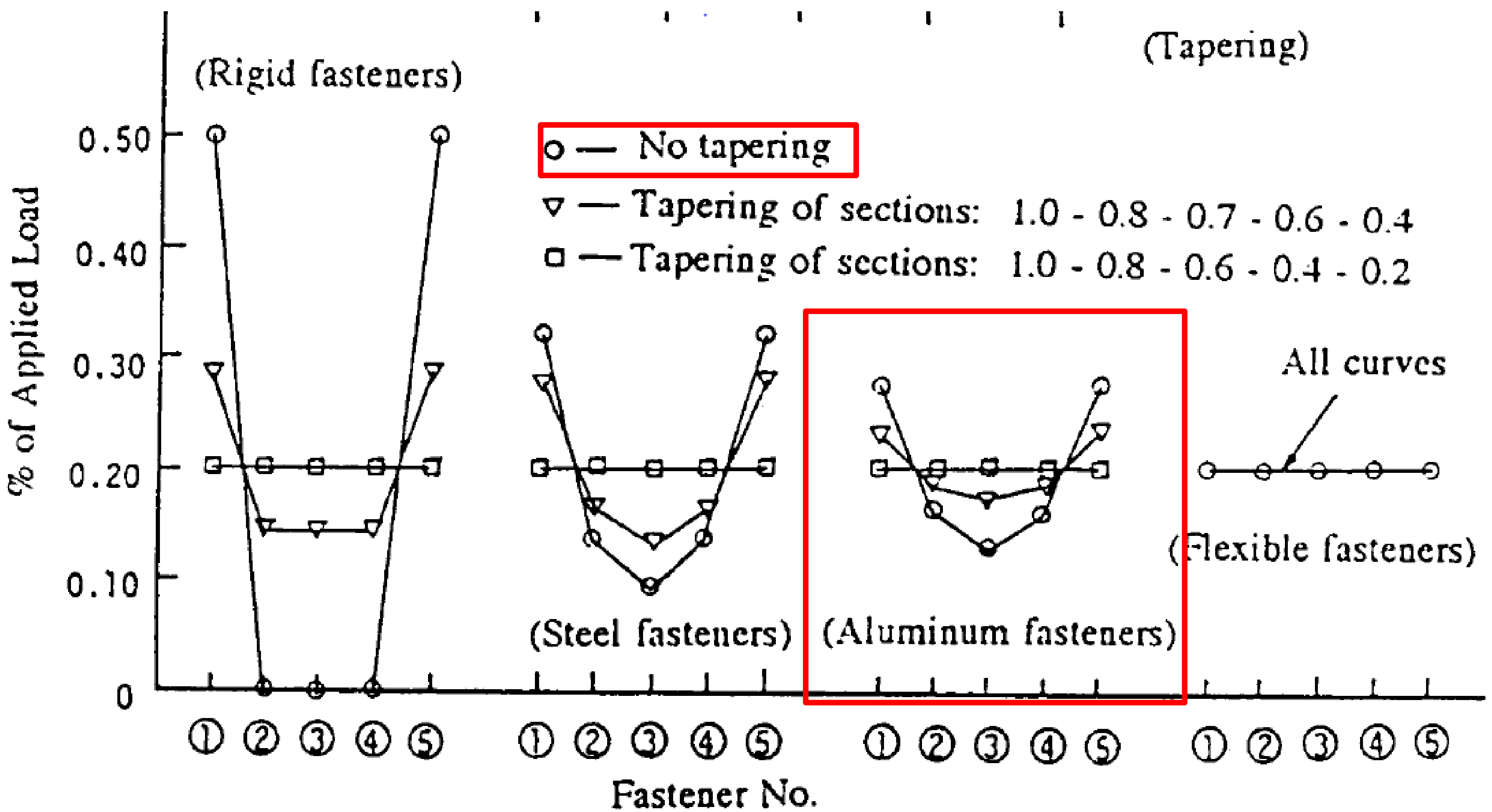


Fig. 9.12.7 Four Fastener Lap Joint

ATTACHMENT FATIGUE ANALYSIS



Load transfer: 30%-15%-10%-15%-30%

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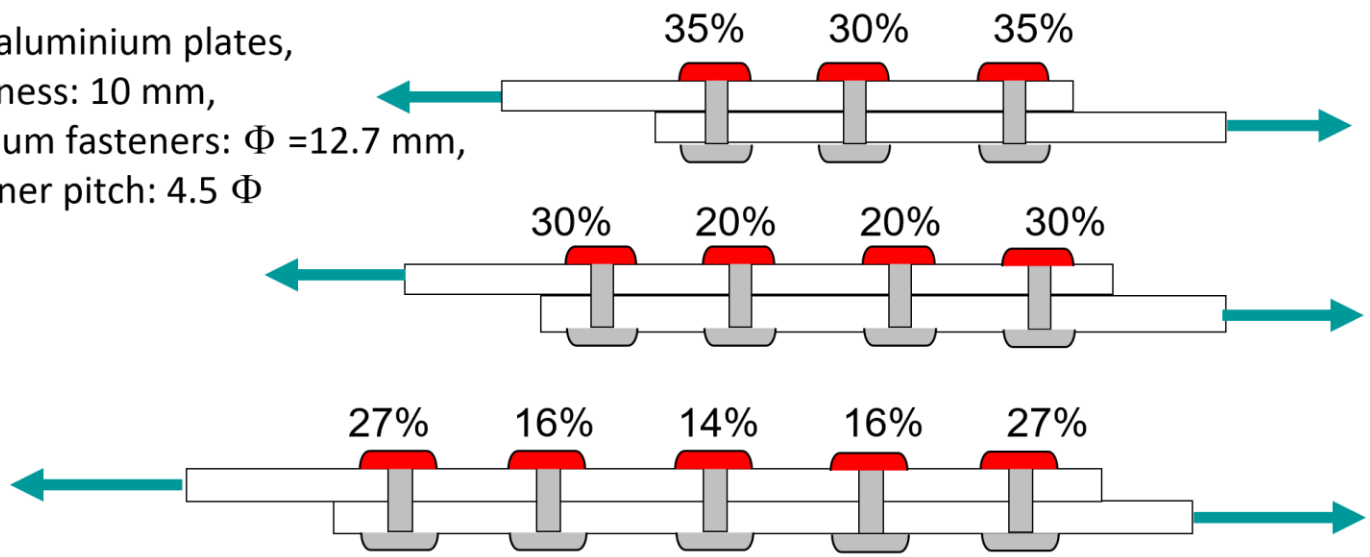
Parameters Affecting Fatigue Endurance

REMINDER

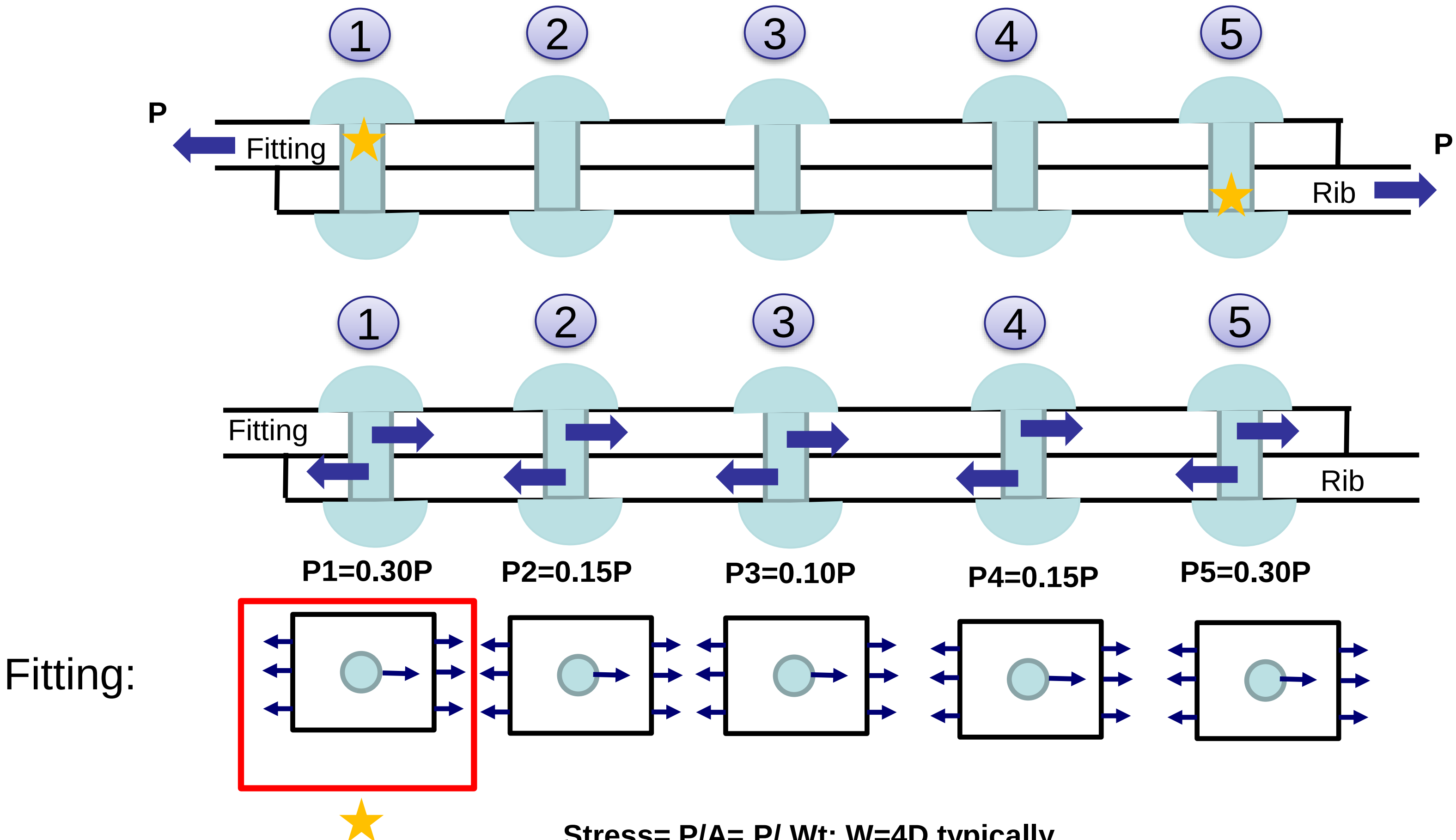
Stress concentration effect – Focusing on fastener joints

Two aluminium plates,
thickness: 10 mm,
titanium fasteners: Φ =12.7 mm,
fastener pitch: 4.5 Φ



6 fasteners: 25% 15% 10%...
10 fasteners: 24% 13% 6% 4% 3% ...
The load transfer in the first fastener tends to an asymptotic value.

The most critical fastener is fastener 1 on the fitting and fastener 5 on the rib, since they have the highest bypass stress and highest load transfer. Both fasteners should be analysed, but in this exercise, due to the symmetry of the problem the result is the same.



Stress= P/A= P/ Wt; W=4D typically

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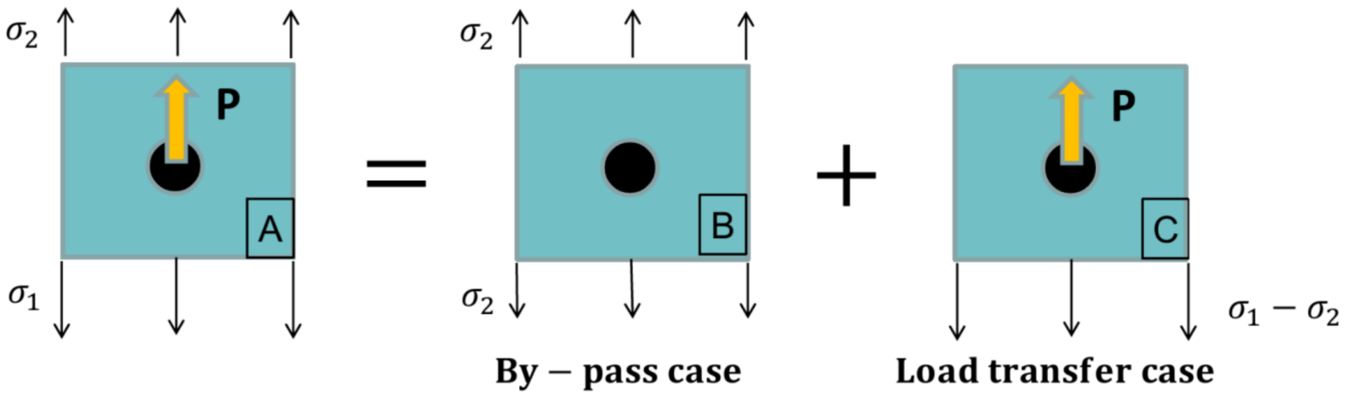
STRESS CONCENTRATION FACTORS

REMINDER

SUPERPOSITION OF LOAD CASES

The stress concentration factor of hole 1 of specimen A can be calculated as superposition of two load cases, by-pass (B) and load transfer (C). The Kt's only can be summed when they are refered to the same stress.



$$K_{tA} = K_{tB} + K_{tC}$$

STRESS CONCENTRATION FACTORS

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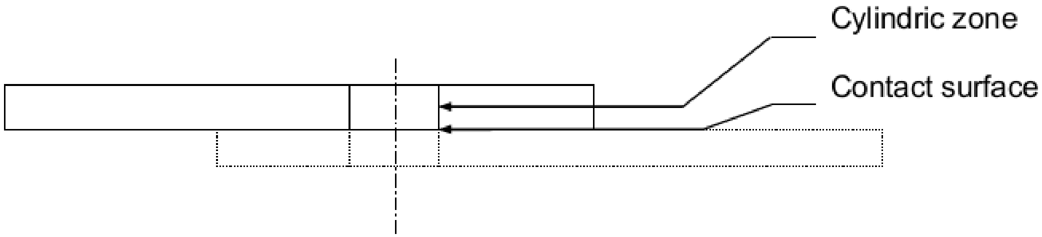
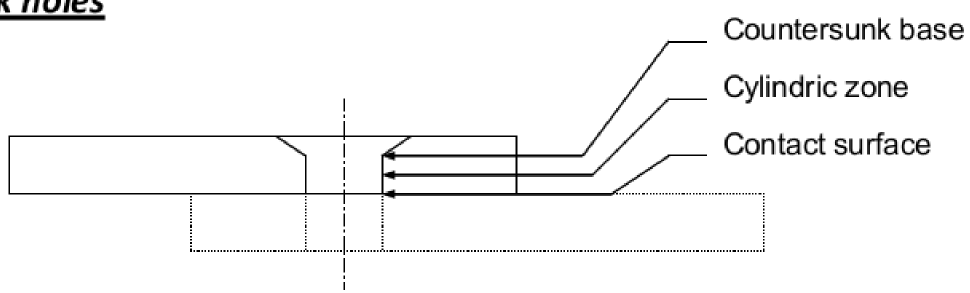
REMINDER

ADDITIONAL EFFECTS OVER Kt.

Cilindrical holes



Countersunk holes



ATTACHMENT FATIGUE ANALYSIS

STRESS CONCENTRATION FACTORS

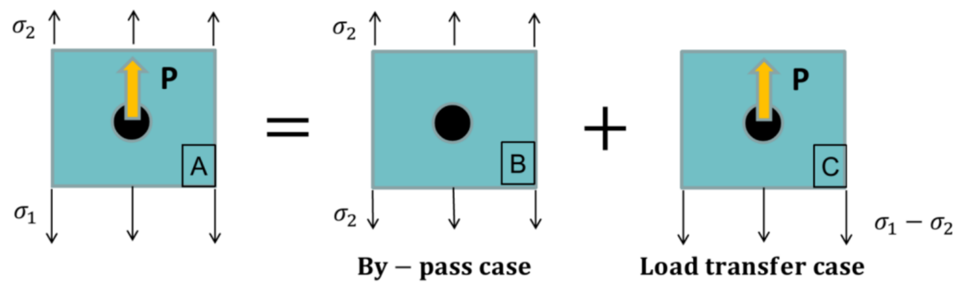
REMINDER

ADDITIONAL EFFECTS OVER Kt.

	Load	Surface	By-Pass load case (By-Pass Kt)				Load transfer case (Bearing Kt)			
			Shadow	Thick.	Counter	Pin bend.	Shadow	Thick.	Counter	Pin bend.
Cylindric hole	Axial	Cylindric Zone	Yes	Yes	No	No	No	Yes	No	No
		Contact Surface	Yes	No	No	No	No	No	No	Yes
	Shear	Cylindric Zone	Yes	Yes	No	No	No	Yes	No	No
		Contact Surface	Yes	No	No	No	No	No	No	Yes
Countersunk hole	Axial	Cylindric Zone	Yes	Yes	No	No	No	Yes	No	No
		Counter. Base	Yes	No	Yes	No	No	No	Yes	No
		Contact Surface	Yes	No	No	No	No	No	No	Yes
	Shear	Cylindric Zone	Yes	Yes	No	No	No	Yes	No	No
		Counter. Base	Yes	No	Yes	No	No	No	Yes	No
		Contact Surface	Yes	No	No	No	No	No	No	Yes

Conservatively, the shadow effect is not considered in the analysis.

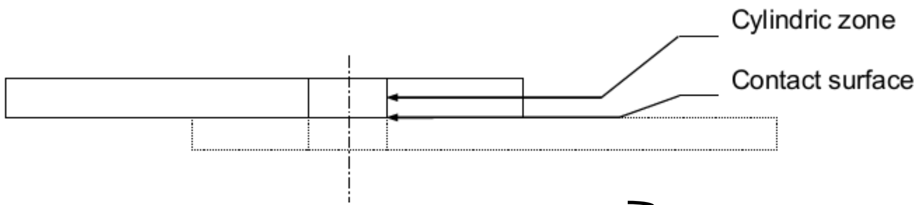
ATTACHMENT FATIGUE ANALYSIS



Without additional effects over Kt:

$$\sigma_{max} = \underbrace{\boxed{Kt_{net} \times \sigma_{net}}}_{Bypass} + \underbrace{\boxed{Kt_{brg} \times \sigma_{brg}}}_{Load\ Transfer}$$

Cilindrical holes



With additional effects over Kt:

$$\sigma_{max,\ cyl} = Kt_t \underbrace{\boxed{Kt_{net} \times \sigma_{net}}}_{Bypass} + Kt_t \underbrace{\boxed{Kt_{brg} \times \sigma_{brg}}}_{Load\ Transfer}$$

$$\sigma_{max,\ cont} = \underbrace{\boxed{Kt_{net} \times \sigma_{net}}}_{Bypass} + Kt_{pb} \underbrace{\boxed{Kt_{brg} \times \sigma_{brg}}}_{Load\ Transfer}$$

$$\sigma_{max} = \max(\sigma_{max,\ cyl}, \sigma_{max,\ cont})$$

$$\sigma_{ref\ net} = \sigma_{1\ net^*}$$

$$Kt_{net} = \sigma_{max} / \sigma_{ref\ net}$$

*SN curves are in net stress and Kt net

REMINDER

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STRESS CONCENTRATION FACTORS

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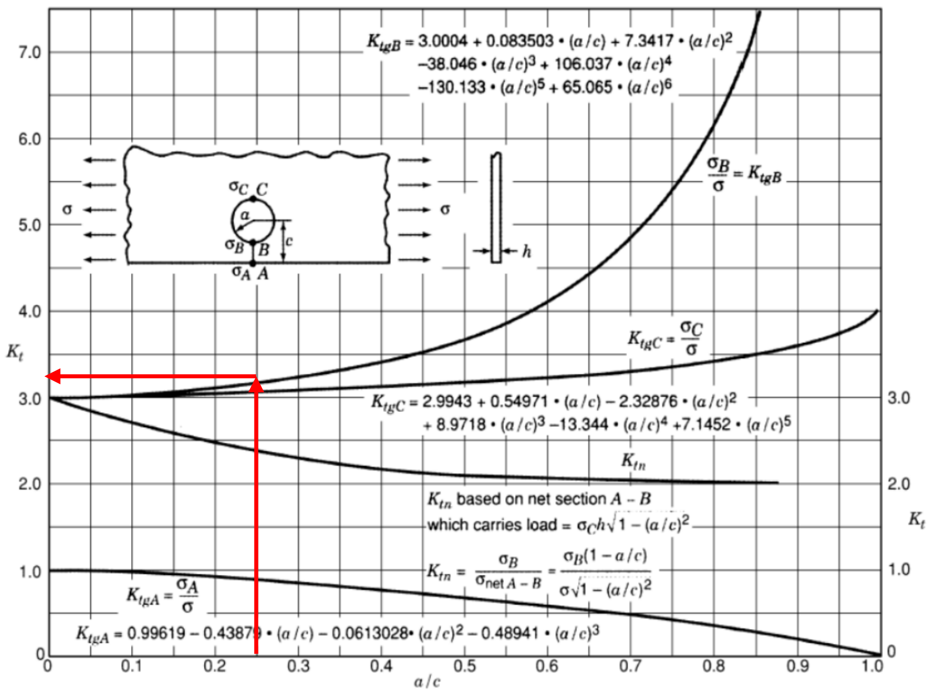
GEOMETRICAL EFFECTS ON THE K_t

EFFECT OF HOLES ON Kt –PROXIMITY OF AN EDGE



Source: Peterson's Stress Concentration Factors (2nd Edition) – © 1997 by John Wiley & Sons

Bypass

$Kt_{net} = ??$

$\sigma_{net} = ??$

Remember:

$Kt_{net} \times \sigma_{net} = Kt_{gross} \times \sigma_{gross}$

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STRESS CONCENTRATION FACTORS

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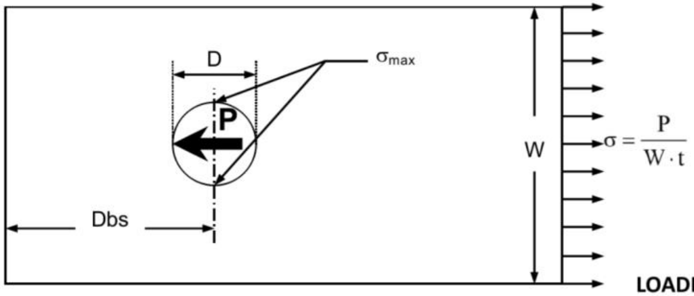
REMINDER

Load transfer

Kt OF HOLES. Loaded hole in finite thin element. Axial load.



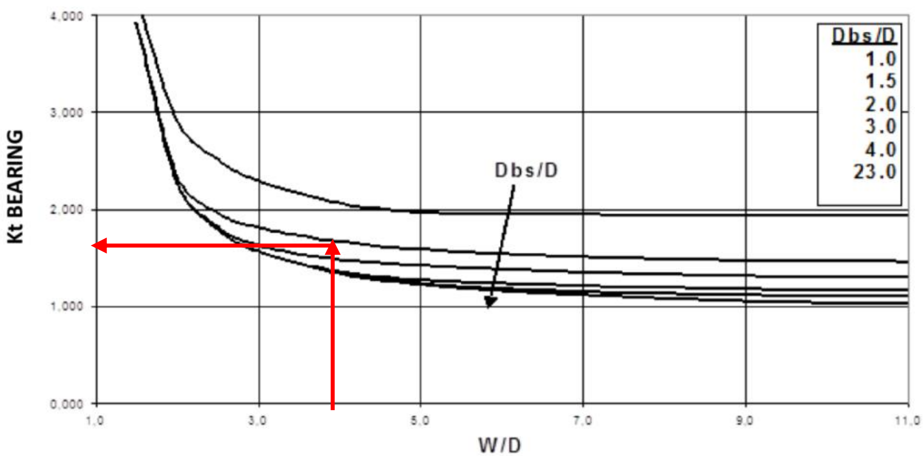
For loaded holes the stress concentration factor is referred to the bearing stress. The maximum stress points are located perpendicularly to the load

$$Kt_{bg} = \frac{\sigma_{max}}{\sigma_{bg}}$$

$$\sigma_{bg} = \frac{P}{D \cdot t}$$

P = Applied load at the hole.
D = Hole diameter.
t = Thickness panel.

LOADED HOLE IN FINITE THIN ELEMENT. AXIAL LOAD.



Kt_{brg}

$\sigma_{ref} = \sigma_{brg} = P/(Dt)$

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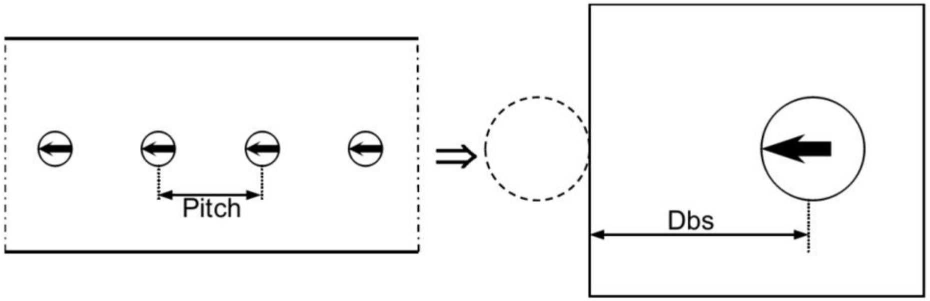
REMINDER

Kt OF HOLES. Loaded hole in finite thin element. Axial load.

In the case of a row of holes all of them with load transfer, the analysis hypothesis is the following:



$$Dbs = Pitch - \frac{D}{2}$$

$$D = Diameter$$

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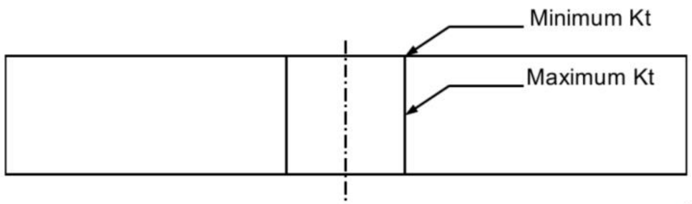
REMINDER

Thickness effect

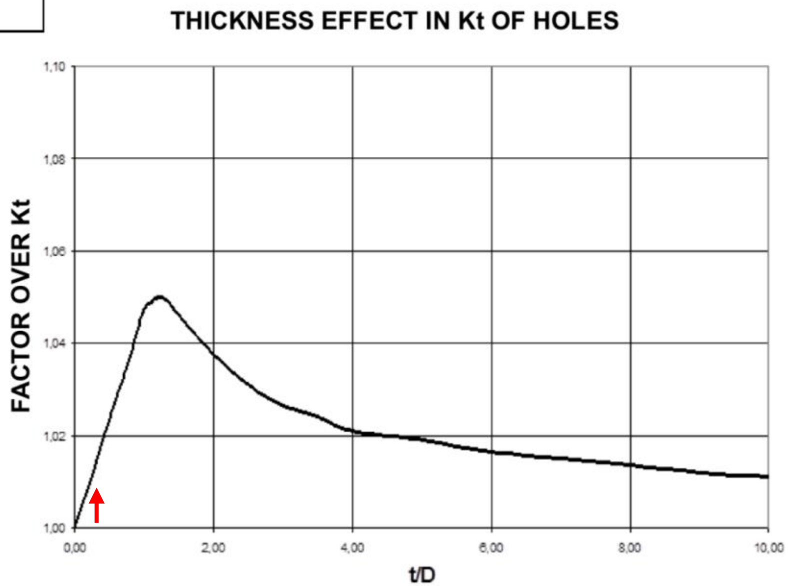
ADDITIONAL EFFECTS OVER Kt.

Thickness effect



Kt_t
Factor over the Kt

The maximum Kt is located in the middle plane, but this effect should not be combined with the fastener bending effect, which produces maximum Kt at the surface.



ATTACHMENT FATIGUE ANALYSIS

STRESS CONCENTRATION FACTORS

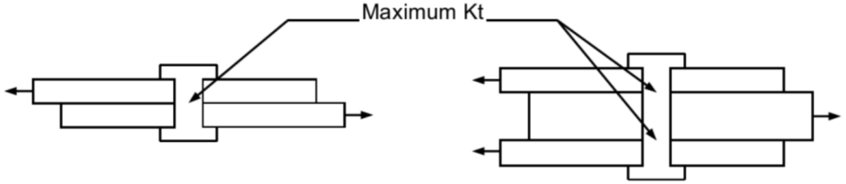
REMINDER

ADDITIONAL EFFECTS OVER Kt.

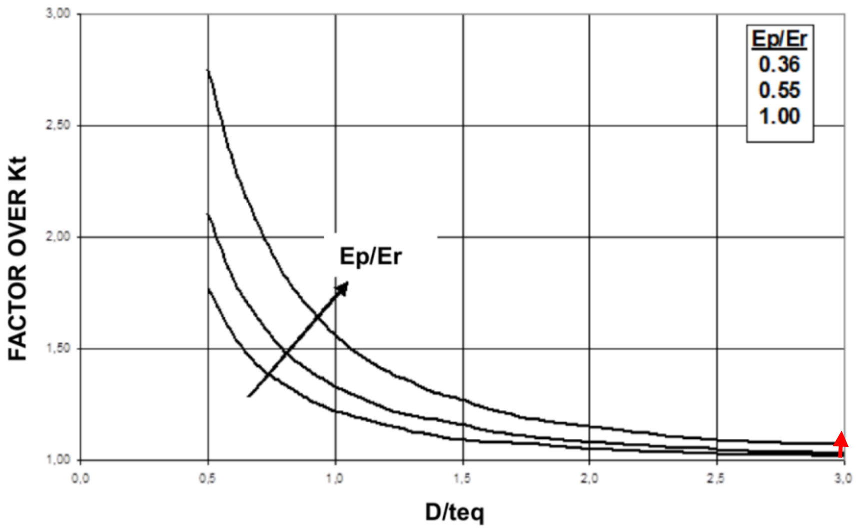
Pin bending effect

The maximum stress is located at contact surface.

This effect must not be combined with thickness effect and countersunk effect which produces maximum Kt at the middle plane).



PIN BENDING EFFECT IN Kt OF OPEN HOLES



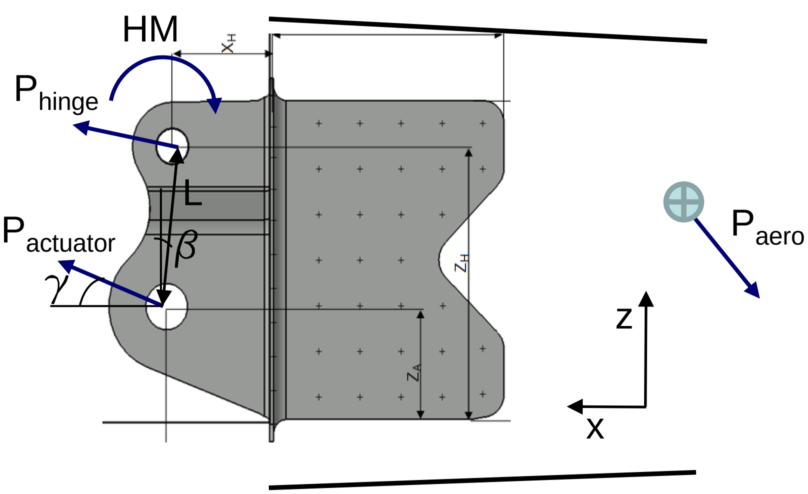
Pin bending effect

Kt_{pb}
Factor over the Kt

1. Get the material and geometry for the location ✓
2. Get the fastener loads and stresses at the joint. Obtain the Kt taking a reference stress. ✓
3. Convert the spectrum to the reference stress
   - ❑ *Get the transfer factor from the hinge moment to load in the fastener row*
   - ❑ *Get the factor from fastener row load to reference stress*
4. Enter in the SN curve to get the fatigue life for the location

ATTACHMENT FATIGUE ANALYSIS

The actuator load was obtained in the lug exercise. The hinge load in each fitting must be obtained considering that the total aerodynamic load is reacted in all the hinges and actuators.

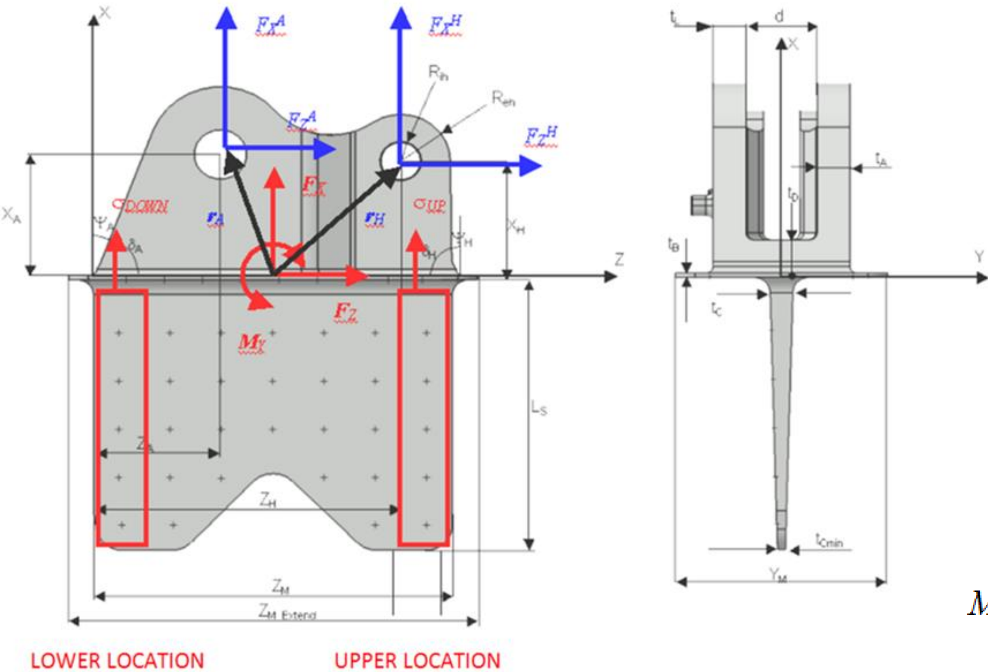


$$P_{aeroX} = P_{HX} + P_{AX}$$
$$P_{aeroZ} = P_{HZ} + P_{AZ}$$

P_{HX}, P_{HZ} Total hinge forces

The total hinge force much be distributed between all the hinges. This could be done manually with simplifications or with FEM (CELAS forces).

ATTACHMENT FATIGUE ANALYSIS



Once the hinge load and the actuator load are obtained, the resultant load and moment are reacted in the attachments.

For the purpose of this exercise we will asume that this joint carries only load in X direction and the load in Z direction is reacted by the web joint.

$$M_Y^{H+A} = \left(F_X^H \cdot a_H - F_Z^H \cdot b_H\right) - \left(F_X^A \cdot a_A + F_Z^A \cdot b_A\right)$$

Where: $\begin{cases} a_H = Z_H - \dfrac{Z_M}{2} \\ b_H = X_H \end{cases}$ and $\begin{cases} a_A = \dfrac{Z_M}{2} - Z_A \\ b_A = X_A \end{cases}$

Conservatively, the 50% of the total F_{UP} and F_{DOWN} is assumed to be carried by the first row of rivets, respectively.

Then $F_{UP} = \dfrac{F_X^H + F_X^A}{2} + \dfrac{M_Y^{H+A}}{a_H + a_A}$ and $F_{DOWN} = \dfrac{F_X^H + F_X^A}{2} - \dfrac{M_Y^{H+A}}{a_H + a_A}$.

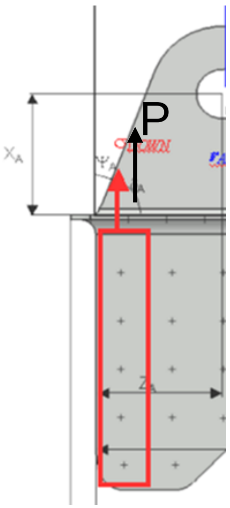
Note: $P_{AX} = F_X^A$, etc...

Note: If a detailed FEM is available with all the rivets, the rivet loads and the stresses can be obtained from it.

Therefore, to get the spectrum in net stress $\sigma_{\mathbf{1net}}$:

1. F_X^A , F_Z^A, F_X^H and F_Z^H must be obtained for all the spectrum lines (max and min)
2. Get the load at the first row of rivets (P), obtained as $0.5F_{UP}$ or $0.5F_{DOWN}$ for each line of the spectrum (max and min)
3. Get the maximum and minimum reference stress for each line:

$$\sigma_{\mathbf{ref}} = \sigma_{1\ net} = P/[(W-D)t]$$



Elevator Angle [deg]	HM max. [kNm]	HM min. [kNm]	Aerodyn. Load max. [kN]	Aerodyn. Load min. [kN]	Occurrences (10000 FC)
5	1.673	-1.911	9.558	-10.92	10000
12	1.207	-1.445	6.897	-8.255	10000
5	0.98	-1.209	5.601	-6.907	15000
3	0.787	-1.01	4.496	-5.771	30000
4	0.785	-1.01	4.488	-5.771	29250
2	0.605	-0.824	3.459	-4.707	58500
1	0.369	-0.585	2.11	-3.345	30000
3	0.315	-0.533	1.798	-3.046	25000
1	0.105	-0.321	0.603	-1.833	331030
1	0.105	-0.321	0.603	-1.833	50000

$\sigma_{1\ net\ max}$, $\sigma_{1\ net\ min}$

ATTACHMENT FATIGUE ANALYSIS

1. Get the material and geometry for the location ✓
2. Get the fastener loads and stresses at the joint. Obtain the Kt taking a reference stress. ✓
3. Convert the spectrum to the reference stress ✓
4. Enter in the SN curve to get the fatigue life for the location
   - *Interpolate if needed*

THANK YOU

for your attention!